

Week 1 – Deliverable Team 10

Data Visualization Associate Internship

Database Setup, Exploration & Data Quality Report

Executive Summary

This report presents the activities completed during **Week 1** of the **Data Visualization Associate Internship**. The primary objective was to establish a complete data analysis environment by configuring PostgreSQL, importing the provided datasets, connecting the database to Supabase and Looker Studio, and performing an initial exploration of the data.

Three datasets—**opportunity**, **learner_opportunity**, and **status_mapping**—were successfully imported into PostgreSQL after resolving data import issues caused by inconsistencies in the source files. The database was then connected to Supabase to enable secure cloud access and integrated with Looker Studio for future dashboard development.

Initial exploratory analysis was performed using SQL queries to understand the structure, completeness, and quality of the datasets. Several data quality issues, including missing values, duplicate records, inconsistent formatting, invalid numeric values, and broken relationships between tables, were identified. These findings establish a strong foundation for the data cleaning, transformation, and visualization tasks planned for the following weeks.

1. Database Setup

1.1 PostgreSQL Database Setup

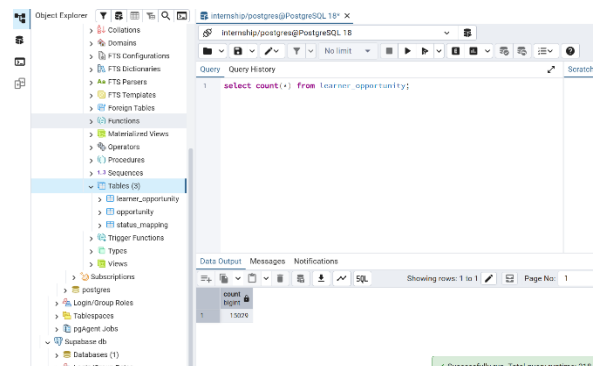
PostgreSQL was set up along with pgAdmin as the management tool. A new database was created, and three tables — opportunity, learner_opportunity, and status_mapping — were created using CREATE TABLE scripts run through the Query Tool. All columns were defined as TEXT type, since the raw data contained inconsistent formats (timestamps written as long numbers, mixed TRUE/FALSE/blank values, embedded JSON fragments), and using TEXT avoided import failures while the data was still being explored.

1.2 Dataset Upload Process

The opportunity and status_mapping datasets were clean and imported successfully on the first attempt using pgAdmin's Import/Export Data feature.

The learner_opportunity dataset failed to import the same way. The source CSV file contained 256 columns per row, while only 43 of those columns had real data — the remaining 213 were empty columns left over from the original Excel export. Since PostgreSQL's import process requires every row to match the target table's column count exactly, this mismatch produced an "extra data after last expected column" error.

To resolve this, the extra empty columns were removed from the file without altering any of the actual data values (verified by comparing the cleaned file against the original row by row). The cleaned file was then imported successfully



using the COPY command run directly in the Query Tool:

```
COPY learner_opportunity
FROM 'C:\full\path\to\learner_opportunity_cleaned_v2.csv'
DELIMITER ','
CSV HEADER;
```

This loaded all 15,029 learner records successfully.

1.3 Connection Method Used: Supabase

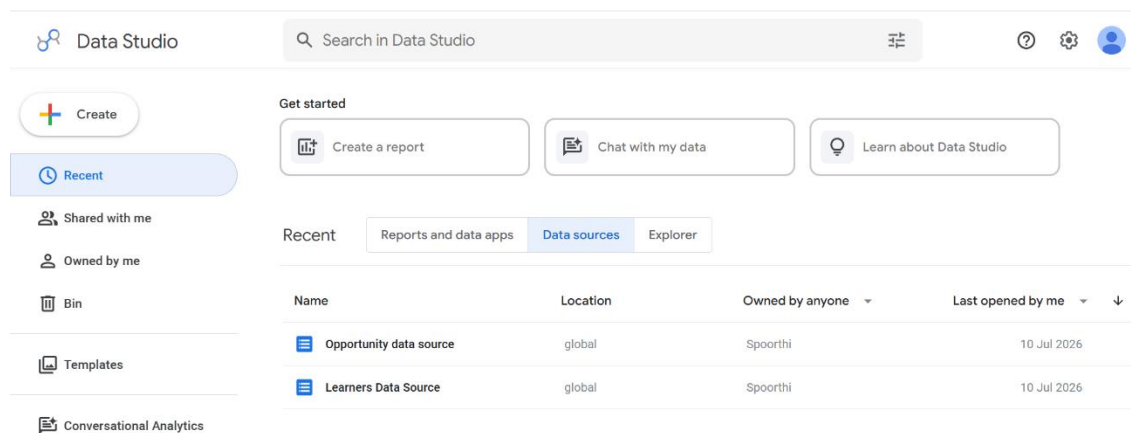
To make the database accessible to Looker Studio, a Supabase PostgreSQL project was created. Supabase provides a publicly reachable Postgres host, which is required since Looker Studio cannot connect to a database running only on a local machine.

Instead of manually recreating each table, a backup of the local database was taken using pgAdmin and restored directly into the Supabase project using pgAdmin's Backup/Restore feature. This copied all three tables and their full data across in a single step, preserving the original structure.

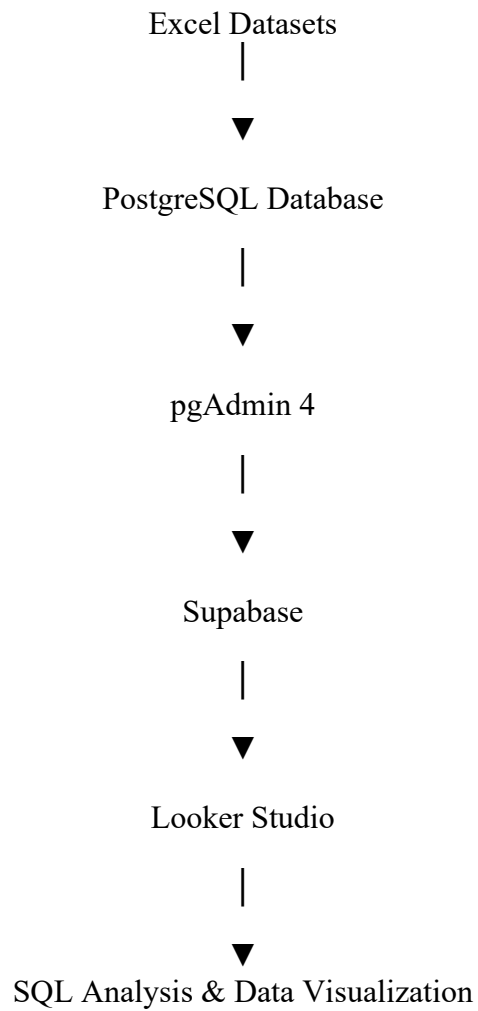
The Supabase database was then connected to Looker Studio using the PostgreSQL connector, with two separate data sources configured using custom SQL queries:

- **Learner Data Source** — joins learner_opportunity with status_mapping so numeric status codes display as readable labels (e.g., 1010 → "Applied").
- **Opportunity Data Source** — the opportunity table connected independently, since no reliable link exists between individual learner applications and specific opportunity records (see Section 4).

```
SELECT
  lo.*,
  sm.status AS status_label
FROM learner_opportunity lo
LEFT JOIN status_mapping sm
  ON lo.status = sm.status_code;
```



Flow Chart:



1.4 Confirmation of Successful Load

Row counts were verified directly in PostgreSQL after import to confirm no data was lost or duplicated:

Table	Expected Records	Verified Records	Status
opportunity	5,733	5,733	Loaded successfully
learner_opportunity	15,029	15,029	Loaded successfully (after cleanup)
status_mapping	14	14	Loaded successfully

1.5 Tools & Technologies Used

The following tools and technologies were used throughout the Week 1 activities:

Tool	Purpose
PostgreSQL 18	Database management system used for storing and querying the datasets
pgAdmin 4	Graphical interface for database creation, table management, SQL execution, and data import
SQL	Used for database creation, querying, validation, and exploratory analysis
Supabase	Cloud-hosted PostgreSQL service used to enable external database connectivity
Looker Studio	Business Intelligence tool used for dashboard creation and data visualization
Microsoft Excel	Used for inspecting source datasets and cleaning the learner dataset before import

2. Dataset Overview

2.1 Table Summary

Table	Records	Fields	Purpose
opportunity	5,733	33	Catalog of opportunities offered (internships, courses, competitions, etc.)
learner_opportunity	15,029	43	Individual learner applications and their progress/status
status_mapping	14	2	Lookup table decoding numeric status codes into readable labels

2.2 Key Fields Identified for Analysis

opportunity table:

- category — type of opportunity (Internship, Career, Competition, Course, Event, Engagement, Masterclass, JobSimulation)
- fee, currency_type — cost structure
- duration, duration_type — length of the opportunity

- location — geographic availability
- Reward, Badge — incentives tied to the opportunity
- is_auto_approve, last_date_to_apply — process/operational fields

learner_opportunity table:

- status — numeric application status code (decoded via status_mapping)
- payment_status, fee, amount_to_be_paid — payment-related fields
- apply_date, completion_date, withdraw_date — application lifecycle timestamps
- work_item_sk — intended reference to the related opportunity (see data quality note in Section 4)
- category — opportunity category as recorded on the application itself

2.3 Initial Observations About Data Structure

- All columns were imported as TEXT type rather than their logical types (numbers, dates, booleans), because the raw data contained inconsistent formatting that would have caused import failures with stricter typing. Type conversion is deferred until after data quality issues are resolved.
- The learner_opportunity file required cleanup before import (213 empty trailing columns from the source Excel export were removed; no actual data values were changed).
- Field completeness varies significantly within both tables — some fields are populated on nearly every record, while others (e.g., summary, Testimonial, DropoutTransaction in the opportunity table) are populated on fewer than 5% of records, suggesting they are deprecated or rarely-used fields.
- The learner_opportunity table includes a work_item_sk field that appears intended to link each application to its corresponding opportunity, but the ID formats between the two tables do not match (see Section 4.4).

3. Exploratory Analysis

The queries below were run directly against the PostgreSQL database to explore patterns, distributions, and relationships across the datasets.

3.1 Opportunity Categories

```
SELECT category, COUNT(*) AS count
FROM opportunity
GROUP BY category
ORDER BY count DESC;
```

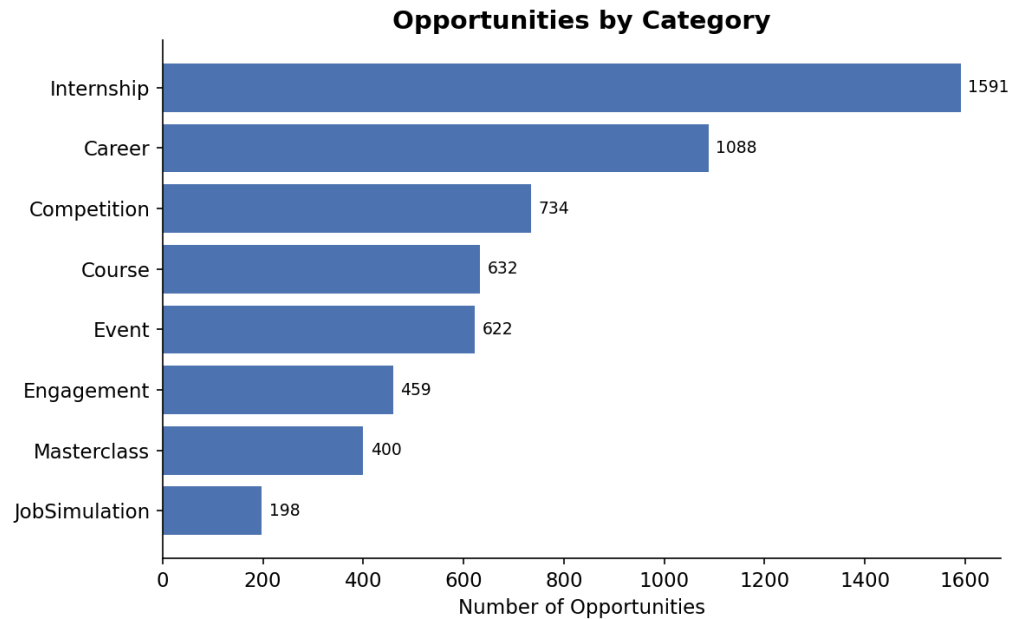


Figure 1: Distribution of opportunities by category

Internship is the most common category (1,591 opportunities), followed by Career (1,088) and Competition (734). JobSimulation is the least common of the eight main categories (198). Two categories, Program and Xploreu, appear only twice each in the dataset — likely newly introduced or test entries rather than established categories. Also there are 4 categories which are not formatted correctly.

Data Output

Messages

Notifications

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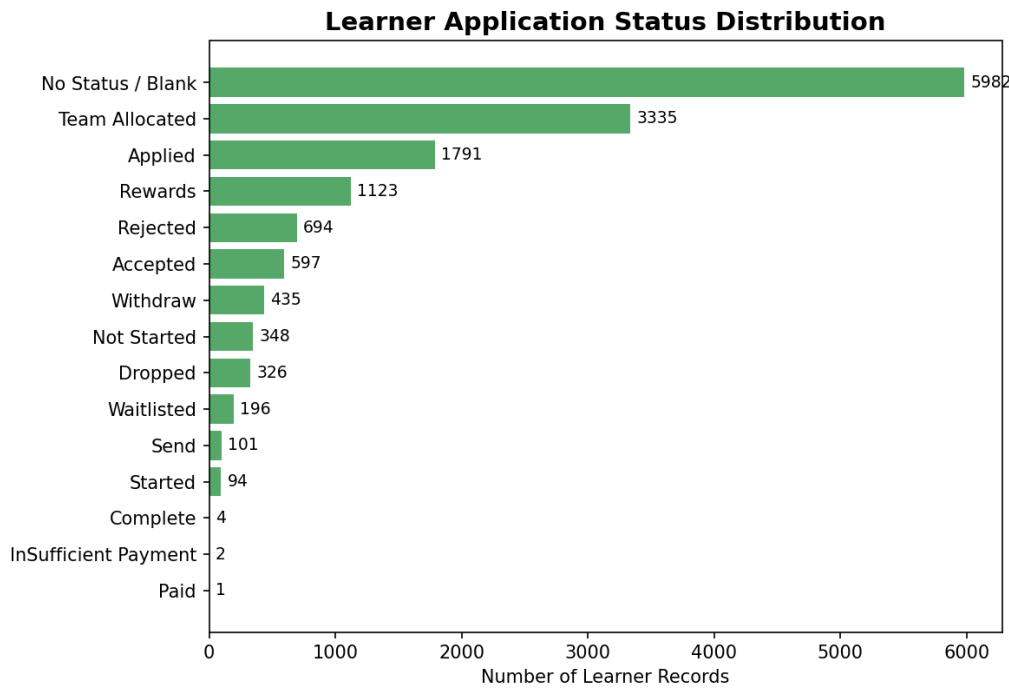
SQL

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	category text	count bigint
1	Internship	1591
2	Career	1088
3	Competition	734
4	Course	632
5	Event	622
6	Engagement	459
7	Masterclass	400
8	JobSimulation	198
9	Program	2
10	Xploreu	2
11	NULL	2
12	{ %22label%22: %22United Kingdom%...	1
13	%22value%22: %221%22 }	1
14	category	1
15	\\ct\\1766057859531\\	1

3.2 Learner Application Status Distribution

```
SELECT sm.status AS status_label, COUNT(*) AS count
FROM learner_opportunity lo
LEFT JOIN status_mapping sm ON lo.status = sm.status_code
GROUP BY sm.status
ORDER BY count DESC;
```



Data Output	Messages	Notifications
status_label text	count bigint	
1	[null]	5982
2	Team Allocated	3335
3	Applied	1791
4	Rewards	1123
5	Rejected	694
6	Accepted	597
7	Withdraw	435
8	Not Started	348
9	Dropped	326
10	Waitlisted	196
11	Send	101
12	Started	94
13	Complete	4
14	InSufficient Paym...	2
15	Paid	1

Figure 2: Learner application status breakdown

The largest single group (5,982 records, about 40% of all learner records) has no status recorded at all — a significant pattern discussed further in Section 4. Among records that do have a status, "Team Allocated" is the most common (3,335), followed by "Applied" (1,791) and "Rewards" (1,123, indicating completed and rewarded applications). "Rejected" (694) and "Withdraw" (435) represent meaningful drop-off points worth investigating further.

3.3 Fee Structure: Free vs Paid Opportunities

```
SELECT
  CASE WHEN fee::numeric = 0 THEN 'Free' ELSE 'Paid' END AS fee_type,
  COUNT(*) AS count
FROM opportunity
WHERE fee ~ '^[0-9]+(\.[0-9]+)?$'
GROUP BY fee_type;
```

Data Output	Messages	Noti
fee_type text	count bigint	
1	Free	4518
2	Paid	1209

Opportunity Fee Structure: Free vs Paid

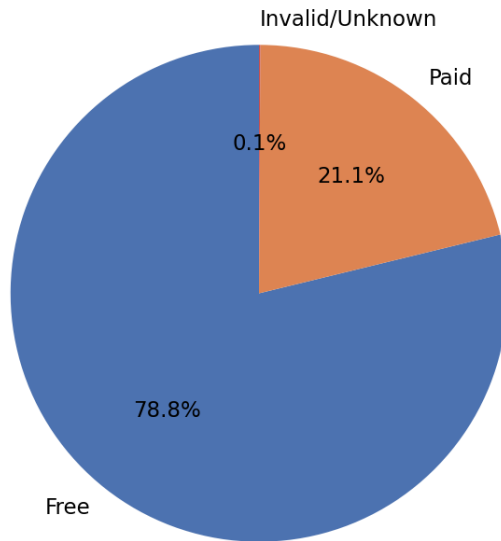


Figure 3: Proportion of free vs paid opportunities

The large majority of opportunities (about 79%) are free, with the remaining 21% carrying a fee. A small number of records (5) had non-numeric or invalid fee values and were excluded from this chart — these are flagged separately in Section 4.

3.4 Top Withdrawal Reasons

```
SELECT withdraw_reason, COUNT(*) AS count
FROM learner_opportunity
WHERE withdraw_reason IS NOT NULL
GROUP BY withdraw_reason
ORDER BY count DESC
LIMIT 10;
```

Data Output

Messages

Notifications

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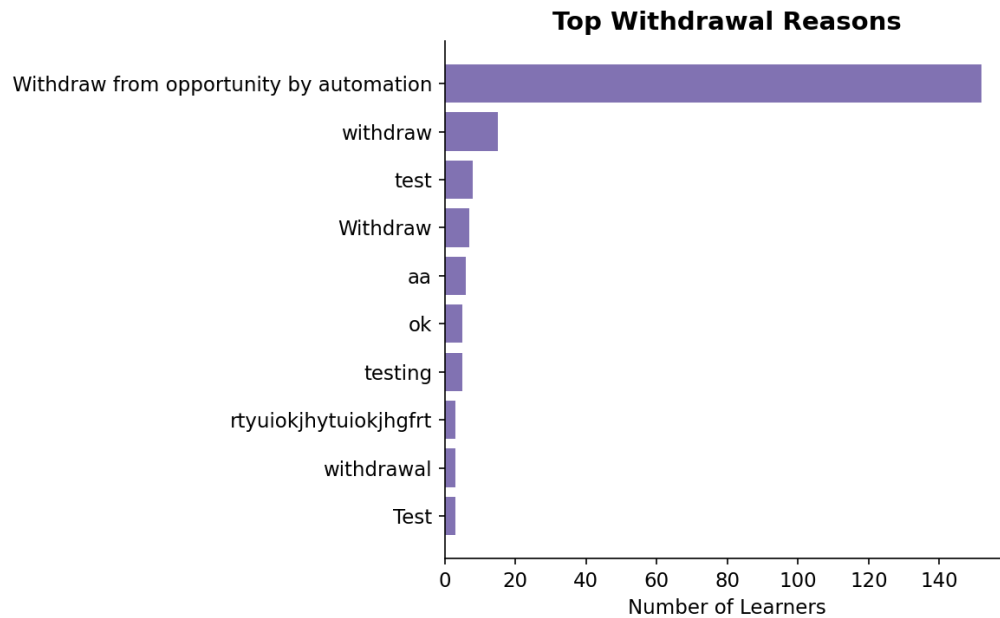


Figure 4: Most common reasons learners withdrew from an opportunity

Only about 2.6% of learner records include a recorded withdrawal reason, but among those that do, clear patterns emerge that could guide retention efforts — this is flagged as an area worth deeper analysis once the underlying status and date fields are cleaned and standardized.

3.5 Opportunities for Deeper Analysis

- Status trends over time — `apply_date`, `completion_date`, and `withdraw_date` are currently stored as raw epoch-millisecond values; converting these to proper timestamps would enable trend analysis (e.g., application volume by month).
- Reward structure analysis — the `Reward` field stores nested reference data (JSON-like) rather than plain values; parsing this would allow a proper reward-distribution analysis.
- Category-level conversion rates — once the `status` field is cleaned, comparing status distribution by opportunity category could reveal which categories have higher acceptance, completion, or dropout rates.
- Duration standardization — the `duration_type` field contains inconsistent values (e.g., "week" vs "weeks") that should be standardized before duration-based analysis.

4. Data Quality Assessment

4.1 Missing Values

Field	Table	% Missing	Note
status	learner_opportunity	~39.8%	Nearly 6,000 records have no application status recorded
work_item_sk	learner_opportunity	~43.7%	No reference to any opportunity
completion_date	learner_opportunity	~92.5%	Very few applications show a recorded completion
withdraw_reason	learner_opportunity	~97.4%	Withdrawal reason rarely captured
summary	opportunity	~96.5%	Field is essentially unused
DropoutTransaction	opportunity	~99.8%	Field is essentially unused

Impact: Fields with very high missingness (summary, Testimonial, DropoutTransaction, NotStartedTransaction) are not reliable for analysis and should be excluded from further work. The high missingness in status and work_item_sk is more serious — it limits how much of the learner dataset can be used for status-based or opportunity-linked analysis.

4.2 Duplicate Records

```
SELECT pk, sk, COUNT(*)  
FROM learner_opportunity  
GROUP BY pk, sk  
HAVING COUNT(*) > 1;
```

The opportunity table has no duplicate opportunity_id values across all 5,733 records — record-level uniqueness is intact there.

The learner_opportunity table, however, contains 187 fully duplicate rows (171 distinct records, each repeated at least once, with every one of the 43 fields identical between copies). This suggests some application records were exported or logged more than once. These duplicates should be removed before any counting-based analysis, since they would otherwise inflate status or category counts artificially.

4.3 Inconsistent Formatting

- duration_type contains inconsistent spelling and pluralization — e.g. "week" (134 records) vs "weeks" (4,153 records), "month" (334) vs "months" (179) — along with at least one typo ("yearssss") and a truncated value ("da"). These need to be standardized before any duration-based grouping is done.
- 19 rows in the opportunity table and 15 rows in the learner_opportunity table contain corrupted values — fragments of URL-encoded JSON text (e.g. %22label%22: %22United Kingdom%22) appearing in fields like category or payment_status where a normal value is expected. This points to a CSV export issue where a free-text field containing special characters broke the row structure during the original export, shifting values into the wrong columns.

- A small number of status codes in learner_opportunity (e.g. '103', '110') do not match any code in status_mapping, and appear to be truncated versions of valid codes (1030, 1010).

4.4 Unusual or Unexpected Values

- The fee field in the opportunity table has a maximum value of over 6 trillion, while the median is 0. This is almost certainly a data entry error rather than a real fee amount, and should be excluded or corrected before using fee in any statistical summary.
- 209 records in learner_opportunity (about 1.4% of the table) have a corrupted pk field — instead of a valid identifier (normally around 44 characters), the field contains a large block of unrelated data, in some cases exceeding 32,000 characters in a single field. This is a serious upstream export or encoding error, separate from the smaller JSON-fragment corruption noted above, and these rows should be excluded from analysis until the cause is understood, since their other fields cannot be trusted either.
- Broken referential integrity between tables: the learner_opportunity table's work_item_sk field uses an ID format prefixed "WorkItem#...", while the opportunity table's opportunity_id field uses a different prefix, "Opportunity#...". Even after stripping the prefixes, none of the 8,342 distinct work_item_sk values match any of the 5,733 opportunity_id values. This means learner applications currently cannot be reliably linked to specific opportunity details (category, fee, location, etc.) using the fields available in these two files.

4.5 Impact on Future Analysis

Taken together, these issues mean that:

- Any analysis relating learner outcomes to opportunity characteristics (e.g., "which categories have the highest acceptance rate") cannot currently be performed directly, since the two tables cannot be joined reliably. Resolving this would likely require either a missing mapping table or a correction to how work_item_sk is generated upstream.
- Status-based analysis should account for the ~40% of records with no status, rather than treating the dataset as fully representative of application outcomes.
- Numeric fields (fee, duration) require validation and outlier handling before being used in aggregate calculations such as averages.
- The 187 duplicate records and 209 records with corrupted identifiers (together roughly 2.6% of the learner dataset) should be filtered out before computing raw counts, since including them would overstate application volume.

5. Conclusion

- Week 1 successfully established the complete data analysis environment required for the internship.
- The PostgreSQL database was configured, datasets were imported and validated, and the database was successfully connected to both Supabase and Looker Studio.
- Initial SQL-based exploration provided valuable insights into the structure and quality of the available data.
- The exploratory analysis identified several important data quality concerns, including missing values, duplicate records, inconsistent formatting, invalid numerical values, and broken relationships between learner and opportunity records.
- These findings will guide the data cleaning and transformation activities in Week 2.
- Overall, the database environment is fully prepared for advanced SQL analysis, dashboard development, and business intelligence reporting in the upcoming phases of the internship.